United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

SET - A

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Develop a JavaScript application where users can track their weekly study hours by subject and get feedback. **[10]**

At first, create two input fields: **Subject Name** and **Study Hours**, an **"Add Study"** button, and a label area to display total study hours, average study hours, subjects studied, and feedback. The user will enter the subject name and the number of hours studied in each session. When the button is clicked, the application will do the following:

a. Set a weekly study goal of **20 hours**.
b. Add the entered study hours to a running total.
c. Store each entered subject name.
d. Count how many study sessions the user has entered.
e. Calculate the average study hour so far:
   Average = Total Study Hours / Number of Sessions
f. Display:
   - total study hours
   - average study hours
   - subjects studied in all sessions
   - feedback message
g. Display feedback based on total study hours:
   - 0–5 → "Start studying regularly!"
   - 6–12 → "Good progress!"
   - 13–19 → "Almost ready!"
   - 20+ → "Weekly goal reached!"
h. If the user enters more than 7 study sessions without reaching 20 hours, display **"Increase your study time!"**

Sample Input	Sample Output
English-2, English-3, Bangla-4, Physics-5	Total: 14 hours, Average: 3.5 hours, Subjects: [English, English, Bangla, Physics], Almost ready!
Math-5, Physics-6, Chemistry-7, English-3	Total: 21 hours, Average: 5.25 hours, Subjects: [Math, Physics, Chemistry, English], Weekly goal reached!
8 sessions, total 15 hours	Total: 15 hours, Average: 1.875 hours, Increase your study time!

2. Write a PHP program for a student academic tracking system. The program should take three inputs from the user via an HTML form: **[10]**

- **credits earned per course**
- **number of courses completed**
- **target graduation credits**.

First, calculate the **total credits earned** by multiplying the credits per course by the number of courses completed. Classify the academic progress using the following conditions:

- **Excellent:** total credits 120 or more
- **Good:** total credits 90 or more but less than 120
- **Average:** total credits 60 or more but less than 90
- **Poor:** total credits less than 60

The program should then compare the total credits earned with the target graduation credits. Finally, display the total credits, the progress category, the target value, and clearly state whether the progress is **above**, **below**, or **exactly at** the target, along with the difference.

Sample:

Credits/ Course	Courses Completed	Target	Total Credits	Progress	Result
4	30	120	120	Excellent	Target met exactly (0 difference)
3	35	100	105	Good	Above target by 5
3	25	90	75	Average	Below target by 15
4	10	60	40	Poor	Below target by 20

3. Create a database named tourism and a table named tourist_spot with the following structure and values: **[10]**

SpotID	SpotName	Region	Category	Rating	EntryFee	VisitorsPerYear
1	Cox's Bazar Beach	CTG	Beach	4.9	0	150000
2	Sundarbans Forest	KHL	Nature	4.3	150	80000
3	Ratargul Forest	SYL	Nature	4.8	100	60000
4	Kuakata Beach	PTU	Beach	4.5	0	90000
5	Shat Gambuj Mosque	BGH	Heritage	4.7	30	70000
6	Paharpur	NWG	Heritage	4.1	30	40000

Now, do the following **(write full PHP–MySQL code):**

1. Display the SpotName, Region and Rating of all spots whose rating is above 4.5 or whose SpotName contains the word 'Beach'. Sort the result by Rating in descending order.
2. Update: for all spots whose rating is between 4.0 and 4.5 (inclusive) and whose EntryFee is greater than 0, increase the Rating by 0.2 and raise the EntryFee by 10%.
3. For each category, show the category name, the number of spots in it, the average rating, and the total visitors per year but only include categories whose average rating is above 4.4. Sort the result by average rating in descending order.
4. For each region, show the region name and its total estimated yearly revenue, calculated as the sum of (EntryFee × VisitorsPerYear) over its spots. Display only those regions whose total estimated revenue exceeds 1,000,000, ordered from highest to lowest.